

Mobile Robot Localization Using Hybrid AI Techniques

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Abstract

This project demonstrates the implementation of important Artificial Intelligence concepts through a Mobile Robot Localization System. The objective is to estimate the robot's position, find an optimal path to the goal, avoid obstacles, and make intelligent decisions under uncertainty.

Different AI techniques are implemented across CO1–CO6 including environment modeling, BFS search, constraint satisfaction, utility-based decision making, Bayesian localization, and hybrid AI reasoning.

The project uses Python data structures such as lists, sets, queues, and state representations. The system generates explainable reasoning traces, validates constraints, updates localization beliefs, and selects optimal actions.

Overall, the project provides practical understanding of core AI concepts and establishes a foundation for advanced Robotics, Machine Learning, Deep Learning, NLP, and Generative AI applications.

Introduction

Mobile Robot Localization is the process of determining the current position of a robot within an environment.

The robot must:

- Identify its location
- Detect obstacles
- Plan a path toward the goal
- Handle uncertainty in sensor readings
- Make intelligent navigation decisions

The project is divided into six Course Outcomes (CO1–CO6), where each module demonstrates a different AI reasoning technique.

Goals

- Model the robot environment using states and obstacles
- Perform pathfinding using BFS search
- Validate environmental constraints
- Implement utility-based decision making
- Perform Bayesian localization
- Build explainable AI reasoning traces
- Develop a hybrid intelligent localization system

Problem Statement

The mobile robot must move from the Start Position (S) to the Goal Position (G) in a grid environment while avoiding obstacles.

Challenges

- Find the shortest path
- Avoid blocked cells
- Handle uncertainty
- Make intelligent decisions

Expected Solution

S → Goal using the safest and shortest path

System Environment

Start Position

(0,0)

Goal Position

(4,4)

Obstacles

(0,3), (1,1), (1,3), (2,1), (3,2), (3,3)

Data Structures Used

- Grid
- Lists
- Sets
- Queue

Output

- Path Generation
- Constraint Validation
- Robot Localization

C01: Environment Modeling

Objective

- Model robot environment

Concepts Used

- PEAS Model
- State Representation
- Obstacles & Goal

Output

- Environment created
- States identified
- Obstacles detected

C02: BFS Search

Objective

- Find shortest path

Concepts Used

- BFS
- Queue
- Visited Set

Output

- Shortest path found
- Goal reached successfully

C03: Constraint Satisfaction

Objective

- Validate path constraints

Concepts Used

- CSP
- Constraint Checking
- Obstacle Validation

Output

- Invalid moves rejected
- Valid path confirmed

C04: Utility-Based Decision Making

Objective

- Select best action

Concepts Used

- Utility Function
- Risk Evaluation
- Decision Logic

Output

- Paths compared
- Best path selected

C05: Bayesian Localization

Objective

- Estimate robot location

Concepts Used

- Bayes Rule
- Probability Inference
- Sensor Data

Output

- Position estimated
- Localization updated

C06: Hybrid AI Localization

Objective

- Integrate all AI techniques

Concepts Used

- BFS Search
- CSP
- Bayesian Reasoning
- Utility Evaluation

Output

- Optimal path generated
- Localization successful
- Goal reached successfully

Performance Analysis

Performance Analysis

CO	Main Technique	Output
CO1	Environment Modeling	State Representation
CO2	BFS	Shortest Path
CO3	CSP	Constraint Validation
CO4	Utility Function	Decision Making
CO5	Bayesian Reasoning	Localization
CO6	Hybrid AI	Integrated Reasoning

Observations

- BFS successfully found the shortest path
- CSP prevented invalid movements
- Utility functions improved decision quality
- Bayesian reasoning handled uncertainty effectively
- Hybrid AI produced the best overall performance

Grid Environment Diagram



+	-	-	+	-	-	+	-	-	+	-	-	+
	S		0		0		X		0			
+	-	-	+	-	-	+	-	-	+	-	-	+
	0		X		0		X		0			
+	-	-	+	-	-	+	-	-	+	-	-	+
	0		X		0		0		0			
+	-	-	+	-	-	+	-	-	+	-	-	+
	0		0		X		X		0			
+	-	-	+	-	-	+	-	-	+	-	-	+
	0		0		0		0		G			
+	-	-	+	-	-	+	-	-	+	-	-	+

CO1 : OUTPUT

```
===== CO1 : Environment Modeling =====
```

GRID:

```
['S', '0', '0', 'X', '0']  
['0', 'X', '0', 'X', '0']  
['0', 'X', '0', '0', '0']  
['0', '0', 'X', 'X', '0']  
['0', '0', '0', '0', 'G']
```

Start State: (0, 0)

Goal State : (4, 4)

Obstacle Positions:

```
(0, 3)  
(1, 1)  
(1, 3)  
(2, 1)  
(3, 2)  
(3, 3)
```

Environment Successfully Modeled

PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code> █

CO2 : OUTPUT

```
===== CO2 : BFS Search =====
```

GRID:

```
['S', '0', '0', 'X', '0']  
['0', 'X', '0', 'X', '0']  
['0', 'X', '0', '0', '0']  
['0', '0', 'X', 'X', '0']  
['0', '0', '0', '0', 'G']
```

Shortest Path Found:

```
[(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
```

Path Length = 9

Goal Reached Successfully

```
PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code> █
```

CO 3: OUTPUT

```
===== C03 : Constraint Validation =====
```

```
GRID:
```

```
['S', '0', '0', 'X', '0']  
['0', 'X', '0', 'X', '0']  
['0', 'X', '0', '0', '0']  
['0', '0', 'X', 'X', '0']  
['0', '0', '0', '0', 'G']
```

```
Checking Path...
```

```
(0, 0) ✓  
(0, 1) ✓  
(0, 2) ✓  
(1, 2) ✓  
(2, 2) ✓  
(2, 3) ✓  
(2, 4) ✓  
(3, 4) ✓  
(4, 4) ✓
```

```
No Constraint Violations
```

```
Valid Path Confirmed
```

```
PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code> █
```

CO4: OUTPUT

```
===== CO4 : Utility Decision =====
```

```
GRID:
```

```
['S', '0', '0', 'X', '0']  
['0', 'X', '0', 'X', '0']  
['0', 'X', '0', '0', '0']  
['0', '0', 'X', 'X', '0']  
['0', '0', '0', '0', 'G']
```

```
Path A:
```

```
[(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 3), (2, 4), (3, 4), (4, 4)]
```

```
Distance = 9
```

```
Risk = 8
```

```
Utility = 83
```

```
-----  
Path B:
```

```
[(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
```

```
Distance = 9
```

```
Risk = 2
```

```
Utility = 89
```

```
-----  
Best Decision = Path B
```

```
PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code> █
```

CO 5: OUTPUT

```
===== C05 : Bayesian Localization =====
```

```
GRID:
```

```
['S', '0', '0', 'X', '0']  
['0', 'X', '0', 'X', '0']  
['0', 'X', '0', '0', '0']  
['0', '0', 'X', 'X', '0']  
['0', '0', '0', '0', 'G']
```

```
Sensor Reading = (2, 2)
```

```
Prior Probability = 0.6
```

```
Sensor Accuracy = 0.8
```

```
Posterior Probability = 0.686
```

```
Estimated Robot Position:
```

```
(2, 2)
```

```
Localization Updated Successfully
```

```
PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code> █
```

CO 6 : OUTPUT

```
===== CO6 : Hybrid Mobile Robot Localization =====

GRID:
['S', '0', '0', 'X', '0']
['0', 'X', '0', 'X', '0']
['0', 'X', '0', '0', '0']
['0', '0', 'X', 'X', '0']
['0', '0', '0', '0', 'G']

STEP 1 : Search
Path Found:
[(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]

STEP 2 : Constraint Checking
Constraints Satisfied = True

STEP 3 : Bayesian Localization
Estimated Position = (2,2)
Belief = 0.686

STEP 4 : Utility Evaluation
Utility = 91

FINAL RESULT
Goal Reached Successfully
Localization Successful
Hybrid AI System Executed
PS C:\Users\jaswa\AppData\Local\Programs\Microsoft VS Code>
```

Result

- Shortest path found successfully
- Constraints validated
- Robot localized accurately
- Best decision selected
- Goal reached successfully

Final Path

$S \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4) \rightarrow (3,4) \rightarrow G$

Conclusion

This project successfully implemented important Artificial Intelligence concepts using a Mobile Robot Localization System. Environment modeling, BFS search, constraint satisfaction, utility-based decision making, Bayesian localization, and hybrid AI reasoning were applied to solve robot navigation problems. The system demonstrated explainable reasoning, obstacle avoidance, uncertainty handling, and intelligent path selection. Overall, the project provided practical understanding of core AI algorithms and established a strong foundation for Robotics, Artificial Intelligence, Machine Learning, Deep Learning, NLP, and Generative AI applications.

The background is a dark blue gradient. It features several faint, light blue technical diagrams. On the right side, there is a large circular gauge with concentric circles and radial lines, resembling a speedometer or a dial. Below it, there are smaller circular diagrams with arrows indicating rotation. On the left side, there are also faint circular elements. The overall aesthetic is technical and modern.

Thank
You